

Sutton and Barton Exercises Chapter 7

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1 Chapter 7

Exercise 7.1 In Chapter 6 we noted that the Monte Carlo error can be written as the sum of TD errors (6.6) if the value estimates don't change from step to step. Show that the n-step error used in (7.2) can also be written as a sum TD errors (again if the value estimates don't change) generalizing the earlier result.

$$\begin{aligned} G_{t:t+n} - V_{t+n-1}(S_t) &= R_{t+1} + \gamma G_{t+1:t+n} - V_{t+n-1}(S_t) + \gamma V_{t+n-1}(S_{t+1}) - \gamma V_{t+n-1}(S_{t+1}) \\ &= \delta + \gamma(G_{t+1:t+n} - \gamma V_{t+n-1}(S_{t+1})) \Rightarrow \\ &\delta + \gamma(G_{t+2:t+n} - \gamma V_{t+n-1}(S_{t+2})) \Rightarrow \\ &\sum_{k=t}^{t+n-1} \gamma^{k-t} \delta_k \end{aligned}$$

Exercise 7.3 Why do you think a larger random walk task (19 states instead of 5) was used in the examples of this chapter? Would a smaller walk have shifted the advantage to a different value of n? How about the change in left-side outcome from 0 to -1 made in the larger walk? Do you think that made any difference in the best value of n?

Because with only 5 states, there are so few states, it is hard to see the difference (and benefit) of using n-step methods. The bias-variance tradeoff is more pronounced with 19 states. To be sure, there is higher bias in fewer step methods and more variance in higher step methods.

A smaller walk would have shifted the advantage to a smaller n. In particular for this problem, n=2 would be best. If n=3, then the value of C could be pulled towards 1, which would be inaccurate.

No, the optimal value of n would not change, as: "Scaling rewards scales the variance up or down, but it doesn't change which n trades off bias vs

variance best.” In other words, the bias-variance structure is maintained. If one wanted to change the optimal value of n , one could: change the transition dynamics, add stochastic resets, change the environment to make early termination more common, or add stochastic reset (randomly get send back to the starting position).

Exercise 7.4 Prove that the n -step return of Sarsa (7.4) can be written exactly in terms of a novel TD error, as (too lazy to write it out)

$$\begin{aligned}
G_{t:t+n} - Q_{t-1}(S_t, A_t) &= R_{t+1} + \gamma G_{t+1:t+n} - Q_{t-1}(S_t, A_t) \Rightarrow \\
G_{t:t+n} - Q_{t-1}(S_t, A_t) &= R_{t+1} + \gamma G_{t+1:t+n} - Q_{t-1}(S_t, A_t) - \gamma Q_t(S_{t+1}, A_{t+1}) + \\
&\quad \gamma Q_t(S_{t+1}, A_{t+1}) \\
&= \delta_t + \gamma(G_{t+1:t+n} - Q_t(S_{t+1}, A_{t+1})) \Rightarrow \\
\delta_t + \gamma \delta_{t+1} + \gamma^2(G_{t+2:t+n} - Q_{t+1}(S_{t+2}, A_{t+2})) &\Rightarrow \\
\sum_{k=t}^{t+n-1} \gamma^{k-t} \delta_k + \gamma^n(G_{t+n:t+n} - Q_{t+n-1}(S_{t+n}, A_{t+n})) & \\
G_{t+n:t+n} - Q_{t+n-1}(S_{t+n}, A_{t+n}) &\text{ is 0, however. We can add } Q_{t-1}(S_t, A_t) \text{ over} \\
&\text{and recover 6.6.}
\end{aligned}$$

Exercise 7.5 Write the pseudocode for the off-policy state-value prediction algorithm described above.

We have to initialize $V(s)$ arbitrarily because we’re working with state values.

Next: If $\tau + n < T$, then: $\rho_\tau(R_{t+1} + \gamma G_{\tau+1:h}) + (1 - \rho_\tau(V_{\tau+n-1}(S_t)))$

Then: $V(S_\tau) \leftarrow V(S_\tau) + \alpha[G - V(S_\tau)]$

Exercise 7.5 Prove that the control variate in the above equations does not change the expected value of the return.

The control variate part is: $\gamma Q_{h-1}(S_{t+1}, A_{t+1}) + \gamma \bar{V}_{h-1}(S_{t+1})$

Taking $\gamma E[Q_{h-1}(S_{t+1}, A_{t+1}) + \bar{V}_{h-1}(S_{t+1})]$

$$= \sum_a \mu(A_{t+1}|S_{t+1}) \frac{\pi(A_{t+1}|S_{t+1})}{\mu(A_{t+1}|S_{t+1})} [Q_{h-1}(S_{t+1}, A_{t+1}) + \bar{V}_{h-1}(S_{t+1})]$$

$$= \sum_a \pi(A_{t+1}|S_{t+1})[Q_{h-1}(S_{t+1}, A_{t+1}) + \bar{V}_{h-1}(S_{t+1})]$$

Because $\gamma \bar{V}_{h-1}(S_{t+1})$ does is not a probability distribution over actions, $\sum_a \pi(A_{t+1}|S_{t+1})\bar{V}_{h-1}(S_{t+1}) = \bar{V}_{h-1}(S_{t+1})$. And now:

$$\sum_a \pi(A_{t+1}|S_{t+1})[Q_{h-1}(S_{t+1}, A_{t+1})] = \bar{V}_{h-1}(S_{t+1}) \text{ and so:}$$

$$\bar{V}_{h-1}(S_{t+1}) - \bar{V}_{h-1}(S_{t+1}) = 0.$$

Exercise 7.7 Show that the general (off-policy) version of the n-step return (7.13) can still be written exactly and compactly as the sum of state-based TD errors (6.5) if the approximate state value function does not change.

Define the TD error:

$$\delta_t = \rho_t(R_{t+1} + \gamma G_{t+1:h} - V(S_t))$$

Subtract $V(S_t)$ so we can form the TD error:

$$G_{t:h} - V(S_t) = \rho_t(R_{t+1} + \gamma G_{t+1:h}) + (1 - \rho_t)V_{h-1}(S_t) - V(S_t)$$

$$= \rho_t(R_{t+1} + \gamma G_{t+1:h} - V(S_t) + (1 - \rho_t)(V(S_t) - V_{h-1}(S_t)))$$

where above we used the fact that $V(S_t) = V(S_t)(1 - \rho_t) + \rho_t V(S_t)$.

Now under the assumption of the approximate state value function not changing, $V(S_t) = V_{h-1}(S_t)$, we get:

$$G_{t:h} - V(S_t) = \rho_t(R_{t+1} + \gamma G_{t+1:h} - V(S_t))$$

Now $G_{t+1:h}$ can be unrolled as:

$$G_{t+1:h} = \rho_{t+1}(R_{t+2} + \gamma G_{t+2:h}) + (1 - \rho_{t+1})V(S_{t+1})$$

which we can substitute back into $G_{t:h} - V(S_t)$:

$$\rho_t(R_{t+1} + \gamma(\rho_{t+1}(R_{t+2} + \gamma G_{t+2:h}) + (1 - \rho_{t+1})V(S_{t+1}) - V(S_t))).$$

For emphasis, note again that V does not change, so $(1 - \rho_{t+1})V(S_{t+1})$ again gets cancelled out, and we get:

$$\rho_t(R_{t+1} + \gamma(\rho_{t+1}(R_{t+2} + \gamma G_{t+2:h}) - V(S_t))).$$

This repeats for all intermediate terms, and the last term $V(S_h)$ also disappears, as we'd have $G_{h:h} - V(S_h)$ (with γ and ρ factors shared between them). We thus ultimately end up with:

$$G_{t:h} - V(S_t) = \sum_t^{t-1} \gamma^{h-t} (\prod_k^{h-1} \rho_k) \delta_t$$

Exercise 7.11 Show that if the approximate action values are unchanging, then the tree-backup return (7.16) can be written as a sum of expectation-based TD errors: $G_{t:t+n} = Q(S_t, A_t) + \sum_{k=t}^{\min(t+n-1, T-1)} \delta_k \prod_{i=t+1}^k \gamma \pi(A_i | S_i)$.

With some clever math and using 7.8, 7.16 can be rewritten as:

$$G_{t:t+n} = R_{t+1} + \gamma(\bar{V}(S_{t+1}) + \pi(A_{t+1}|S_{t+1})(G_{t+1} - Q(S_{t+1}, A_{t+1})))$$

and then subtract $Q(S_t, A_t)$ to get:

$$G_{t:t+n} - Q(S_t, A_t) = R_{t+1} + \gamma\bar{V}(S_{t+1}) + \pi(A_{t+1}|S_{t+1})(G_{t+1} - Q(S_{t+1}, A_{t+1})) - Q(S_t, A_t)$$

$$= (R_{t+1} + \gamma\bar{V}(S_{t+1}) - Q(S_t, A_t)) + \gamma\pi(A_{t+1}|S_{t+1})(G_{t+1} - Q(S_{t+1}, A_{t+1}))$$

$$= \bar{\delta}_t + \gamma\pi(A_{t+1}|S_{t+1})(G_{t+1} - Q(S_{t+1}, A_{t+1}))$$

Now take into account the term $(G_{t+1} - Q(S_{t+1}, A_{t+1}))$ – we can expand as before:

$$G_{t+1} - Q(S_{t+1}, A_{t+1}) = \bar{\delta}_{t+1} + \gamma\pi(A_{t+2}|S_{t+2})(G_{t+2} - Q(S_{t+2}, A_{t+2}))$$

and substitute back into the original express to get:

$$\begin{aligned} G_{t:t+n} - Q(S_t, A_t) &= \bar{\delta}_t + \gamma\pi(A_{t+1}|S_{t+1})(\bar{\delta}_{t+1} + \gamma\pi(A_{t+2}|S_{t+2})(G_{t+2} - Q(S_{t+2}, A_{t+2}))) \\ &= \bar{\delta}_t + \gamma\pi(A_{t+1}|S_{t+1})(\bar{\delta}_{t+1}) + \gamma^2(\pi(A_{t+1}|S_{t+1})\pi(A_{t+2}|S_{t+2}))(G_{t+2} - Q(S_{t+2}, A_{t+2})). \end{aligned}$$

this ultimately yields the original equation:

$$G_{t:t+n} = Q(S_t, A_t) + \sum_{k=t}^{\min(t+n-1, T-1)} \delta_k \prod_{i=t+1}^k \gamma \pi(A_i | S_i).$$